

House Price Prediction – Project Summary

Internship Project, Week 1 | Shivapuja Sai Kiran Goud | B.Tech CSE (Data Science), HITAM, Hyderabad

Objective

The goal of this project was to build a regression model that predicts house prices based on property features such as area, number of bedrooms and bathrooms, number of stories, and amenities like air conditioning and parking. The dataset used was the Housing Prices Dataset from Kaggle, containing 545 houses and 13 columns, with price as the target variable. The data had no missing values or duplicate rows, so the main cleaning step was converting categorical yes/no columns and furnishing status into numeric form using one-hot encoding before training the models.

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	9,70,043	13,24,507	0.653
Random Forest Regressor	10,21,546	14,00,566	0.612

Training samples: 436 | Testing samples: 109 | Split: 80% train / 20% test. Linear Regression slightly outperformed Random Forest on this dataset, with a lower error and a higher R² score — a reasonable outcome given the dataset has only 545 rows, which isn't always enough for a more complex model like Random Forest to show its advantage.

Which features influence house price the most?

Based on the correlation heatmap, area, number of bathrooms, and air conditioning have the strongest relationship with price — larger houses with more bathrooms and AC consistently sell for more. The number of bedrooms had a noticeably weaker influence on price than expected.

How accurate was the model, in plain terms?

The Linear Regression model achieved an R² score of about 0.65, meaning it explains roughly 65% of the variation in house prices using the available features. This is a decent but not perfect fit — the remaining variation likely comes from factors not captured in the dataset, such as exact location, neighborhood, or house condition.

What surprised me in the data?

Linear Regression performed slightly better than Random Forest here (0.653 vs 0.612 R²), even though Random Forest is usually considered the stronger model. This is likely because the dataset only has 545 rows, which isn't quite enough data for Random Forest's added complexity to pay off.

Recommendation

For a real estate business, this suggests that pricing and marketing strategies should emphasize area, bathroom count, and amenities such as air conditioning, since these are the strongest price drivers buyers actually pay for — rather than over-indexing on bedroom count alone.

Tools used: Python, Pandas, Scikit-learn, Matplotlib, Seaborn | Dataset: Housing Prices Dataset (Kaggle, 545 rows, 13 columns)